

Project Title: Automatic street light using LDR.

Project Overview:

This project is an automatic street light system that uses an LDR (Light Dependent Resistor) to detect surrounding light intensity. The circuit automatically turns the light ON during darkness and turns it OFF during daylight, helping to save energy and reducing the need for human intervention.

Objective

The objective of this project is to design and simulate an automatic lighting system using an LDR sensor. The system demonstrates how sensors can be used for automatic control applications.

Components Used:

1. Breadboard.
2. 9V Battery
3. Transistor(NPN BJT-BC547)
4. LED
5. Potentiometer
6. Photoresistor/LDR(Light Dependent Resistor)
7. Resistor
8. Jumper Wires.

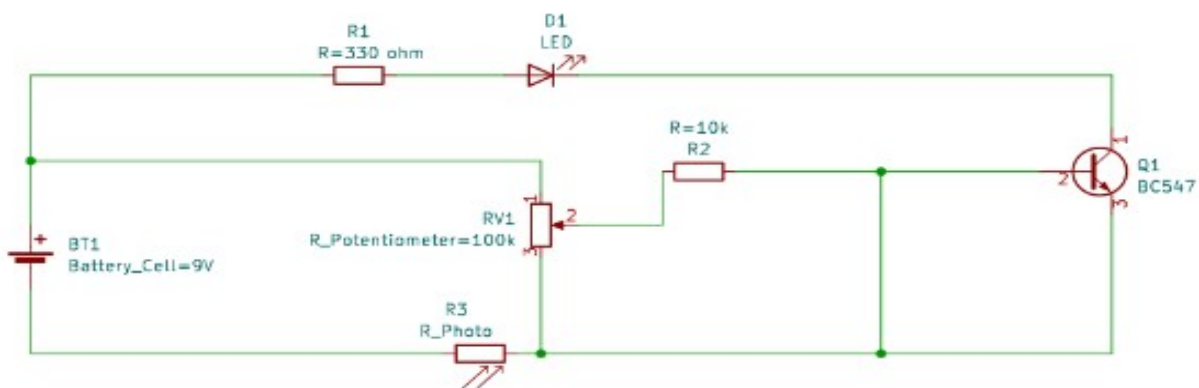


Figure: KiCad Schematic Design of Automatic street light using LDR.

Design Tools Used

- KiCad (Schematic Design)
- Tinkercad (Circuit Simulation)
- GitHub (Project Repository)

Circuit Description:

The circuit starts from the positive terminal of the 9V battery, which provides power to the entire system. The circuit is divided into two main branches. The first branch contains a 330Ω resistor connected in series with an LED to limit the current and protect the LED. The second branch contains a 100kΩ potentiometer and a photoresistor (LDR), which form a voltage divider connected between the positive supply and ground.

The adjustable center wiper of the potentiometer provides a control voltage through a 10k Ω resistor to the base of the BC547 transistor. The transistor works as an electronic switch, allowing current to flow through the LED when the LDR detects darkness. When enough light is present, the transistor remains OFF and the LED stays OFF

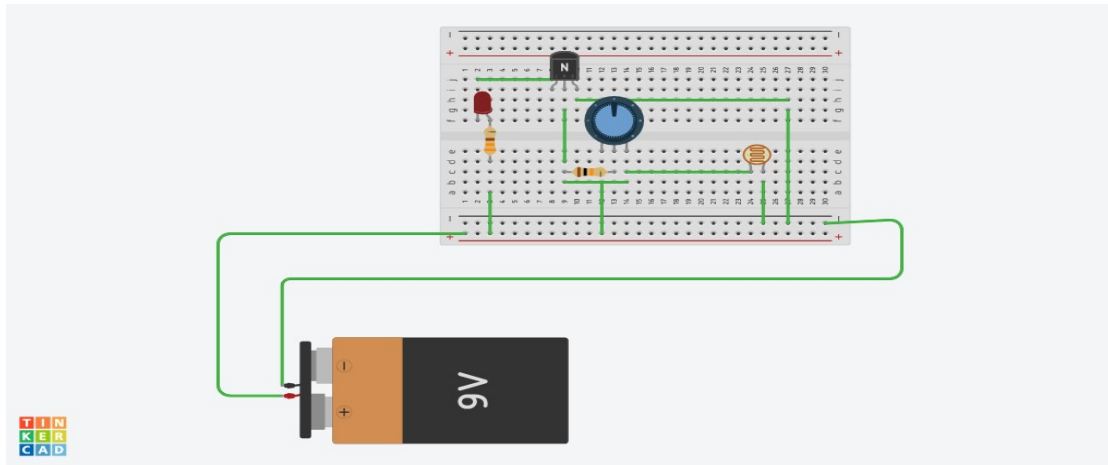


Figure: Tinkercad Simulation of Automatic street light using LDR

Working Principle

1. **Light Sensing:** The potentiometer and photoresistor (LDR) form a voltage divider across the 9V power supply, producing a voltage that changes with light intensity.
2. **Switching Trigger:** The change in LDR resistance controls the transistor's base voltage, turning it ON or OFF.
3. **Activating the LED:** When the transistor turns ON in darkness, current flows through the LED, causing it to illuminate.

Result

The automatic street light circuit was successfully designed and simulated. The LED responds automatically according to the surrounding light conditions.

Conclusion

This project demonstrates the practical use of an LDR sensor for automatic lighting control and shows how electronic circuits can be designed using KiCad and tested using Tinkercad.